Linear Algebra: Advanced Concepts

Part 2

Exploring linear independence, basis and dimension, coordinate systems, kernel and image, and affine spaces. Building on foundational concepts to master advanced linear algebra theory and applications.

Roadmap

Part 2: Advanced Concepts

(5) Linear Independence

Understanding when vectors cannot be expressed as linear combinations of each other.

(6) Basis and Rank

Minimal spanning sets and the dimension of vector spaces and matrices.

(7) Linear Mappings

Kernel, image, rank-nullity theorem, and coordinate transformations.

(8) Affine Spaces

Vector spaces shifted from the origin and affine transformations.

Linear Independence

Fundamental Concept

DEFINITION

Vectors $v_1, v_2, \ldots, v_n$ are **linearly independent** if the only solution to:

$$\lambda_1 v_1 + \lambda_2 v_2 + \ldots + \lambda_n v_n = 0$$

is $\lambda_1 = \lambda_2 = \ldots = \lambda_n = 0$

Intuition

No vector can be written as a linear combination of the others. Each vector contributes new information to the set.

Contrapositive

Vectors are **linearly dependent** if there exist coefficients (not all zero) such that their linear combination equals zero.

Geometric Interpretation

In $\mathbb{R}^2$, two vectors are independent if they don't point in the same direction. In $\mathbb{R}^3$, three vectors are independent if they don't lie in the same plane.

Examples

Independent vs. Dependent

Linearly Independent

Standard Basis in $\mathbb{R}^3$

$e_1 = (1,0,0),\ e_2 = (0,1,0),\ e_3 = (0,0,1)$

The only way to get the zero vector is if all coefficients are zero.
$\lambda_1 e_1 + \lambda_2 e_2 + \lambda_3 e_3 = (\lambda_1, \lambda_2, \lambda_3) = 0$

Polynomials

$P_2 = \{1, x, x^2\}$

No polynomial is a multiple of another. They describe fundamentally different curves.

Linearly Dependent

Collinear Vectors

$v_1 = (1, 2),\ v_2 = (2, 4)$

v_2 is a scalar multiple of v_1.
$2v_1 - v_2 = 0$ (Non-trivial solution exists)

Redundant Vector

$v_1 = (1,0),\ v_2 = (0,1),\ v_3 = (1,1)$

v_3 can be formed by adding v_1 and v_2.
$v_3 = v_1 + v_2$

Spanning Sets

Generating Vector Spaces

DEFINITION

The **span** of a set of vectors $\{v_1, \ldots, v_n\}$ is the set of all possible linear combinations of these vectors.

$$\text{span}(v_1, \ldots, v_n) = \{ \lambda_1 v_1 + \ldots + \lambda_n v_n \mid \lambda_i \in \mathbb{R} \}$$

Generating Set

If $\text{span}(v_1, \ldots, v_n) = V$, we say that the set $\{v_1, \ldots, v_n\}$ **generates** or **spans** the vector space V. Every vector in V can be built from these building blocks.

Geometric Intuition

- **1 Vector:** Spans a line through the origin.
- **2 Vectors:** Span a plane through the origin (if independent).

Basis

Structure of Vector Spaces

DEFINITION

A set of vectors $B = \{b_1, \ldots, b_n\}$ is a basis of a vector space V if it satisfies two conditions simultaneously.

1. Linear Independence

No vector in the set can be written as a linear combination of the others.

"No redundancy."

2. Spanning Property

Every vector in V can be written as a linear combination of vectors in B.

"Enough to cover the space."

Key Insight

A basis is a **minimal spanning set** and a **maximal linearly independent set**. It provides a unique coordinate system for V.

Basis Properties

Structure & Uniqueness

UNIQUENESS OF REPRESENTATION

If B is a basis for V, then every vector $v \in V$ can be written in exactly one way as a linear combination of the basis vectors.

$$v = \lambda_1 b_1 + \ldots + \lambda_n b_n$$

Minimal Spanning Set

A basis is the **smallest** set of vectors that still spans the entire space V.

If you remove any vector from the basis, the remaining set no longer spans V.

Maximal Independent Set

A basis is the **largest** set of linearly independent vectors in V.

If you add any vector to the basis, the set becomes linearly dependent.

Dimension

Measuring Vector Spaces

DEFINITION

The dimension of a vector space V, denoted **dim(V)**, is the number of vectors in any basis of V.

$$\dim(V) = |B|$$

BASIS THEOREM

All bases of a finite-dimensional vector space have the **same number** of elements.

Independence Limit

If dim(V) = n, any set of more than n vectors in V must be linearly dependent.

Spanning Requirement

If dim(V) = n, any set of fewer than n vectors cannot span V.

Dimension Examples

Common Vector Spaces

EUCLIDEAN SPACE $\mathbb{R}^n$

STANDARD BASIS

$$e_1 = (1, 0, \ldots, 0)$$
$$e_2 = (0, 1, \ldots, 0)$$
$$\ldots$$
$$e_n = (0, 0, \ldots, 1)$$

DIMENSION

n

POLYNOMIALS P_k

MONOMIAL BASIS

$$\{ 1, x, x^2, \ldots, x^k \}$$

NOTE

Degree k polynomials have k+1 coefficients (including the constant term).

DIMENSION

k + 1

MATRICES $\mathbb{R}^{m \times n}$

STANDARD MATRIX BASIS

E_{ij} has 1 at (i,j) and 0 elsewhere

COUNT

There are m rows and n columns, so m × n independent positions.

DIMENSION

m × n

Rank of a Matrix

Measuring Linear Independence

DEFINITION

The rank of a matrix A, denoted **rk(A)**, is the dimension of the vector space generated by its columns (the column space).

$$\mathrm{rk}(A) = \dim(\mathrm{Im}(A))$$

Fundamental Theorem

For any matrix A, the number of linearly independent columns equals the number of linearly independent rows.

Column Rank = Row Rank

Linear Mapping Context

If A is the transformation matrix of a linear map Φ, then the rank of A is the dimension of the image of Φ.

It tells us the "dimension of the output".

Rank Properties

Fundamental Theorems

ROW VS. COLUMN RANK

$$\mathrm{rk}(A) = \mathrm{rk}(A^T)$$

The number of linearly independent rows equals the number of independent columns — a key matrix property.

FULL RANK & INVERTIBILITY

$$\mathrm{rk}(A) = n \iff \det(A) \neq 0$$

For a square matrix A (n×n), A is invertible iff it has full rank. If rank < n, it is singular.

RANK OF PRODUCTS

$$\mathrm{rk}(AB) \leq \min(\mathrm{rk}(A), \mathrm{rk}(B))$$

Multiplication cannot increase rank; the product's rank is bounded by the smaller factor's rank.

RANK & SOLUTIONS

$$\mathrm{rk}(A) = \mathrm{rk}(A|b)$$

A system Ax = b has a solution iff the rank of A equals the rank of the augmented matrix (A|b).

Coordinate Systems

The Language of Vectors

THE IDEA

A basis acts like a reference frame. It allows us to describe any abstract vector using a list of numbers (coordinates).

UNIQUENESS

Because a basis is linearly independent and spans the space, the coordinate representation is **unique** for every vector.

THE COORDINATE MAPPING

$$x = \alpha_1 b_1 + \ldots + \alpha_n b_n \quad \Longleftrightarrow \quad [x]_\beta = (\alpha_1, \ldots, \alpha_n)^T$$

The vector **α** is the coordinate vector of x relative to basis B.

Coordinate Representation

From Abstract to Concrete

LINEAR COMBINATION

$$x = \alpha_1 b_1 + \dots + \alpha_n b_n$$

For a fixed basis $B = (b_1, \dots, b_n)$, any vector x can be uniquely expressed as a sum of basis vectors scaled by coefficients α_i.

The Coordinate Vector

We collect the coefficients into a column vector α, called the coordinate of x with respect to B.

$$[x]_\beta = (\alpha_1, \dots, \alpha_n)^T$$

Basis Dependency

Coordinates are **relative**. The same abstract vector x will have different coordinate values if we choose a different basis.

"Coordinates are the language; the basis is the dictionary."

Basis Change Formula

Converting Coordinates

THE SETUP

Consider a vector space V with two ordered bases:
B = (b_1, ..., b_n) and **B' = (b'_1, ..., b'_n)**.
We want to convert coordinates from B to B'.

THE THEOREM

$$[x]_{B'} = T^{-1}[x]_B$$

where T is the transition matrix from B' to the standard basis (or B).

Matrix Construction

The matrix T is formed by placing the basis vectors of B' as columns (if working in R^n) or expressing B' in terms of B.

Inverse Operation

Note the inverse T^{-1}. To go from the "new" basis B' back to "old" basis B, we multiply by T. To go B → B', we multiply by T^{-1}.

Basis Change Example

Concrete Calculation in $\mathbb{R}^2$

1. DEFINE BASES

B (Standard): $e_1=(1,0), e_2=(0,1)$

B' (New): $b'_1=(2,1), b'_2=(1,2)$

3. Transition Matrix T

Columns are b'_1, b'_2 expressed in B.

$$T = [[2, 1], [1, 2]]$$

2. INPUT VECTOR

Vector x has coordinates in B:

$$[x]B = (4, 2)^T$$

4. CALCULATE INVERSE T^{-1}

$\det(T) = 2(2) - 1(1) = 3$

$$T^{-1} = \tfrac{1}{3}\ [[2, -1], [-1, 2]]$$

5. TRANSFORM COORDINATES

$[x]B' = T^{-1}\,[x]B$

$$= \tfrac{1}{3}\ [[2, -1], [-1, 2]] * [4, 2]^T$$
$$= \tfrac{1}{3}\ [6, 0]^T$$

FINAL RESULT IN BASIS B'

$$[x]B' = (2, 0)^T$$

Verification: $2(2,1) + 0(1,2) = (4,2)$ ✓

Transformation Matrix

Representing Linear Maps

For a linear mapping $\Phi: V \to W$ with bases B and C, the **transformation matrix A** relates the coordinate vectors.

$$[\Phi(x)]_C = A\,[x]_B$$

CONSTRUCTION RECIPE

To find the j-th column of A:

1. Take the j-th basis vector of V: $\mathbf{b_j}$
2. Apply the mapping: $\boldsymbol{\Phi(b_j)}$
3. Find its coordinates in W's basis C: $\boldsymbol{[\Phi(b_j)]_C}$

$$A = [\ [\Phi(b_1)]_C \mid \dots \mid [\Phi(b_n)]_C\ ]$$

DIMENSIONS

If dim(V) = n and dim(W) = m:

Matrix A is **m × n**

(m rows, n columns)

Basis Change

General Transformation

THE GENERAL FORMULA

$$\tilde{A}_\varphi = T^{-1} A_\varphi S$$

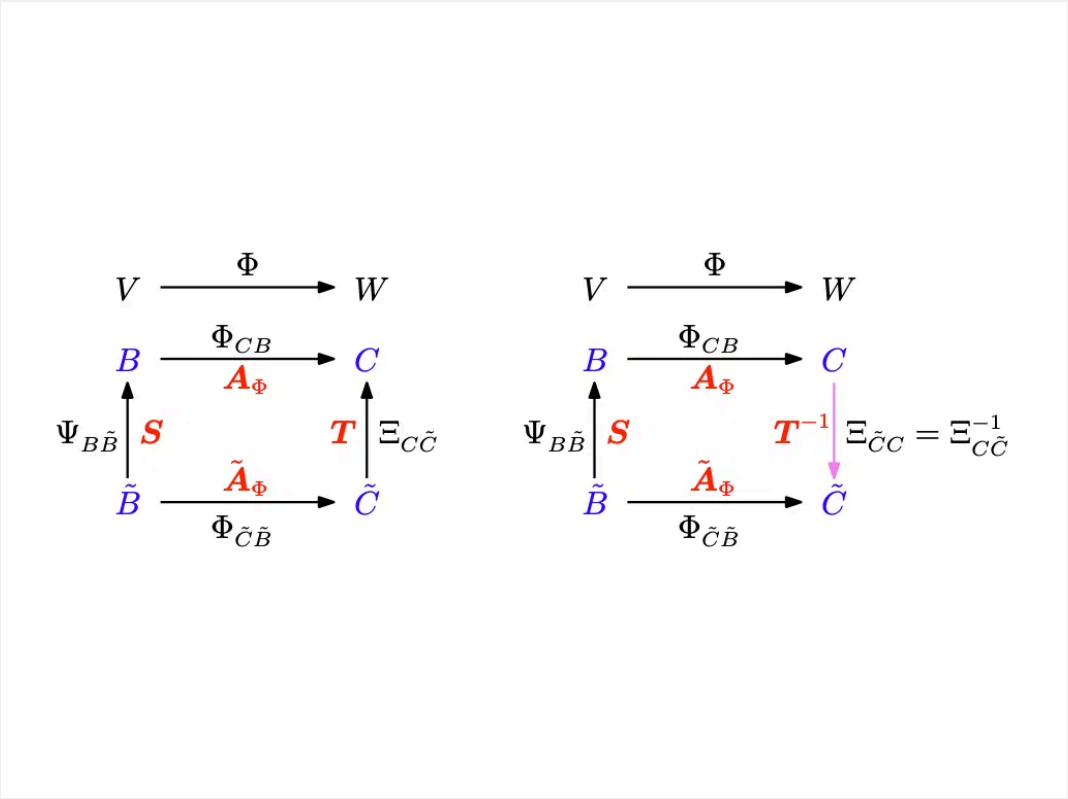
Relates the transformation matrix in the new bases ($\tilde{A}_\varphi$) to the old bases (A_φ).

S : Basis change matrix for V ($B \to \tilde{B}$)

T : Basis change matrix for W ($C \to \tilde{C}$)

A_φ : Matrix of Φ wrt bases B, C

$\tilde{A}_\varphi$: Matrix of Φ wrt bases $\tilde{B}, \tilde{C}$



Commutative diagram showing the relationship between linear maps and basis transformations.

Kernel (Null Space)

The Zero Mapping

DEFINITION

For a linear mapping $\Phi: V \to W$, the kernel is the set of all vectors in V that map to the zero vector in W.

$$\ker(\Phi) = \{ v \in V \mid \Phi(v) = 0_W \}$$

Matrix Interpretation

If $\Phi(x) = Ax$, the kernel is the **null space** of matrix A.

It is the set of all solutions to the homogeneous equation **$Ax = 0$**.

Injectivity Test

The kernel tells us if the mapping is one-to-one.

Φ is injective $\iff$ $\ker(\Phi) = \{0\}$

(Only the zero vector maps to zero).

Image (Range)

The Reach of the Map

DEFINITION

For a linear mapping $\Phi: V \to W$, the image is the set of all possible output vectors in W.

$$\mathrm{Im}(\Phi) = \{ \Phi(v) \mid v \in V \}$$

Matrix Interpretation

If $\Phi(x) = Ax$, the image is the **column space** of matrix A.

It is the span of the columns of A: **span(a_1, ..., a_n)**.

Surjectivity Test

The image tells us if the mapping covers the entire target space.

Φ is surjective $\iff$ $\mathrm{Im}(\Phi) = W$

(Every vector in W is "hit" by the map).

Kernel & Image Properties

Subspaces and Behavior

SUBSPACE THEOREM

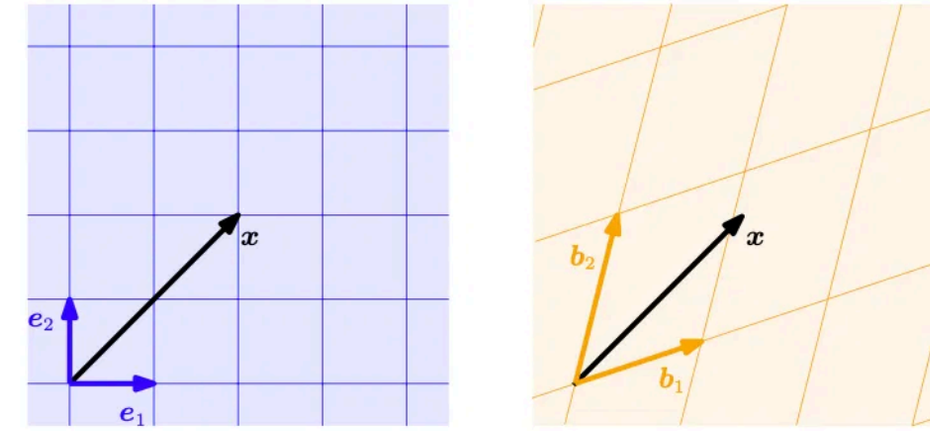
Ker(Φ) is a subspace of the domain V.
Im(Φ) is a subspace of the codomain W.

They are closed under addition and scalar multiplication.

INJECTIVITY TEST

A linear map Φ is injective (one-to-one) if and only if its kernel is trivial.

ker(Φ) = {0} ⟺ Injective



Matrix Connection

When we represent Φ with a matrix A (using coordinates as shown above):

- **Im(Φ)** corresponds to the **Column Space** of A.
- **Ker(Φ)** corresponds to the **Null Space** of A.

Rank-Nullity Theorem

Fundamental Theorem of Linear Maps

THE THEOREM

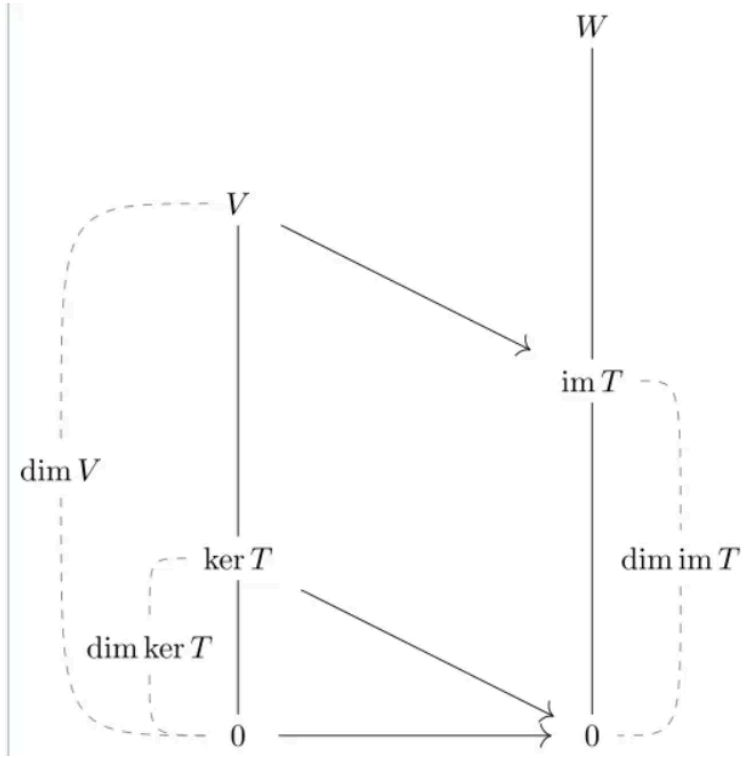
For any linear mapping Φ: V → W, the dimension of the domain V splits into two parts:

$$\dim(V) = \dim(\ker \Phi) + \dim(\operatorname{Im} \Phi)$$

Conservation of Dimension

Think of the input dimension as a resource. Some of it is "lost" (mapped to zero in the kernel), and the rest "survives" (forms the image).

Matrix Form: (Number of Columns) = (Nullity) + (Rank)



Visual representation of the relationship between domain, kernel, and image dimensions.

Rank-Nullity Applications

Dimensional Analysis

"FAT" MATRIX

dim(V) > dim(W) (More columns than rows)

The kernel cannot be trivial (dim ker > 0).
Conclusion: Never injective. Infinite solutions if any exist.

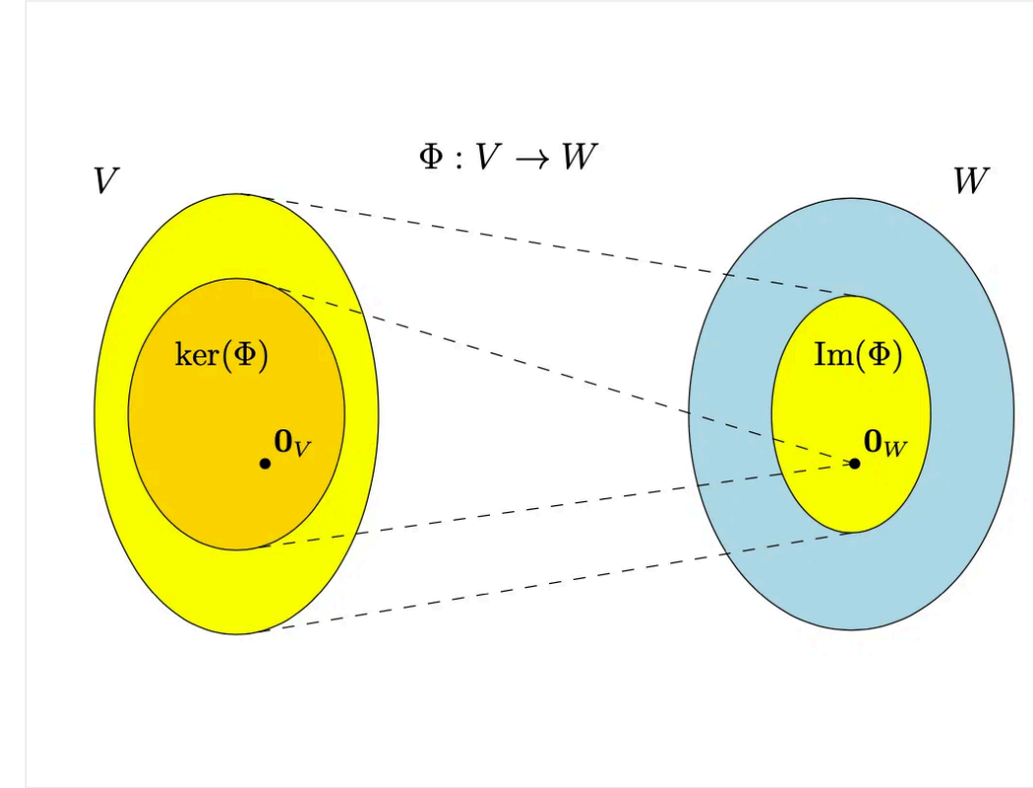
"TALL" MATRIX

dim(V) < dim(W) (More rows than columns)

The image cannot cover W (dim Im < dim W).
Conclusion: Never surjective. No solution for some inputs.

SQUARE MATRIX

dim(V) = dim(W)



The interplay between Kernel (yellow, left) and Image (yellow, right) determines the mapping's properties.

Linear vs. Affine

Clarifying Terminology

LINEAR FUNCTION

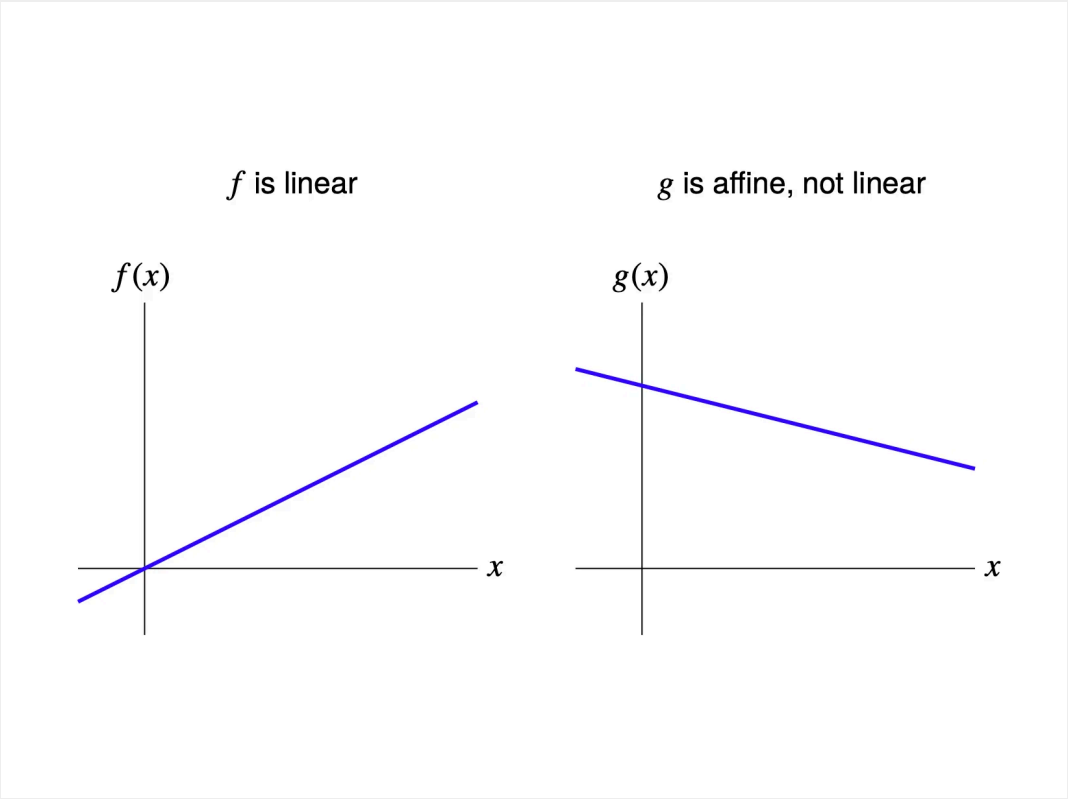
$f(x) = Ax$

- Passes through the origin ($f(0) = 0$).
- Preserves vector addition and scalar multiplication.

AFFINE FUNCTION

$g(x) = Ax + b$

- Does **not** pass through origin (if $b \neq 0$).
- Is a linear transformation followed by a translation.
- Commonly called "Linear Layer" in Neural Networks (terminology clash).



Visual comparison: Linear functions pass through the origin; Affine functions are offset.

Affine Subspace

Shifting Vector Spaces

DEFINITION

An affine subspace L is a vector subspace U that has been shifted by a vector x_0.

$$L = x_0 + U$$

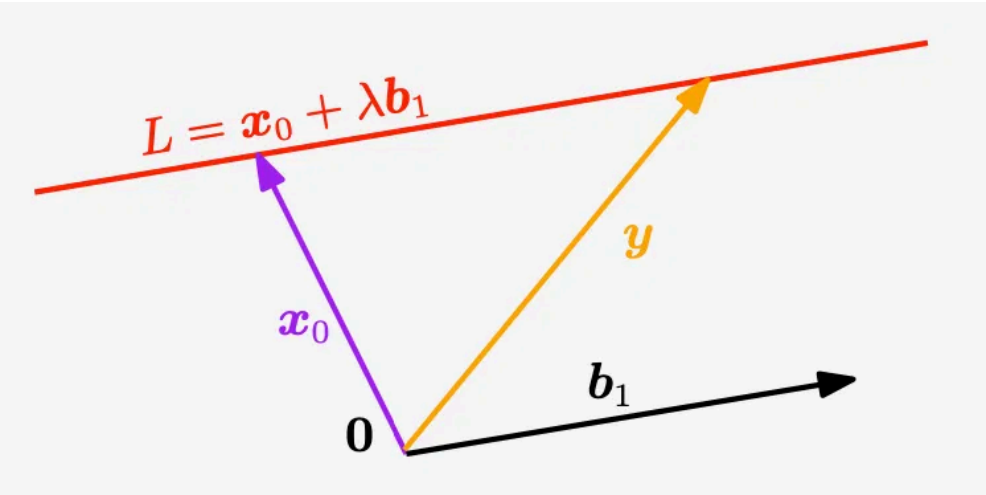
$\{ x_0 + u \mid u \in U \}$

Support Point (x_0)

The "offset" vector. It determines the position of the subspace in space.

Direction Space (U)

The underlying linear subspace. It determines the shape and orientation.



Visualizing an affine line L: a direction vector b_1 shifted by x_0.

Affine Subspace Properties

Anatomy of a Shifted Space

THE STRUCTURE

$L = x_0 + U$

An affine subspace L is formed by translating a linear subspace U (the direction space) by a support vector x_0.

UNIQUENESS

U is unique, x_0 is not

For a given L, the direction space U is unique. However, **any** point in L can serve as the support vector x_0.
$L = y + U$ for any $y \in L$.

DIMENSION

$\dim(L) := \dim(U)$

The dimension of an affine subspace is defined as the dimension of its underlying direction space.
(e.g., a line is 1D, a plane is 2D).

PARAMETRIC EQUATION

$x = x_0 + \lambda_1 b_1 + \ldots + \lambda_k b_k$

Points in L are generated by adding linear combinations of the basis of U to the support point x_0.

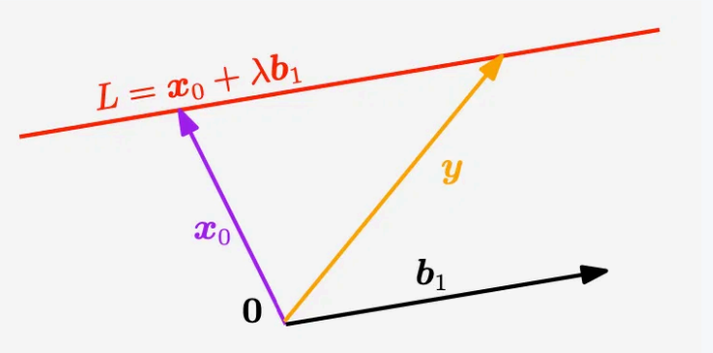
Affine Subspace Examples

From Lines to Hyperplanes

1. THE LINE (1D)

$L = x_0 + \text{span}(u)$

A line in space that does not necessarily pass through the origin.
Parametric: $x(\lambda) = x_0 + \lambda u$



2. THE PLANE (2D)

$P = x_0 + \text{span}(u, v)$

A flat sheet shifted away from the origin.

Parametric:
$x(\lambda, \mu) = x_0 + \lambda u + \mu v$

3. SOLUTION SETS (GENERAL)

$S = \{ x \mid Ax = b \}$

The solution set to a system of linear equations is an affine subspace.

Structure: S = x_part + ker(A)
(Particular solution + Homogeneous solutions)

Affine Mappings

Linear Transformation + Translation

DEFINITION

$$\Phi(x) = Ax + b$$

A Linear Transformation (Matrix)

b Translation Vector (Bias)

KEY PROPERTIES

- Maps affine subspaces to affine subspaces (e.g., lines to lines).
- Preserves ratios of distances along parallel lines.
- Composition of affine maps is affine.

Deep Learning Connection

In Neural Networks, a "Fully Connected Layer" or "Linear Layer" is mathematically an **affine mapping**.

$$y = Wx + b$$

Here, the weights **W** correspond to the linear part **A**, and the bias **b** is the translation.

(Affine maps shift the origin, allowing the model to fit data not centered at zero)

Summary

Key Takeaways

Basis & Dimension

A basis is a minimal generating set. The number of vectors in a basis is invariant and defines the dimension of the space.

Coordinates & Change of Basis

Vectors are represented by coordinates relative to a basis. Changing the basis transforms these coordinates via invertible transition matrices.

Linear Mappings

Represented by matrices. The transformation matrix depends on the choice of basis in both domain and codomain.

Kernel & Image

Kernel: Measures injectivity (one-to-one).
Image: Measures surjectivity (onto).
Both are fundamental subspaces.

Rank-Nullity Theorem

The fundamental conservation law of linear algebra:
dim(Domain) = dim(Kernel) + dim(Image)

Affine Spaces

Generalization of vector spaces that do not necessarily pass through the origin. Defined as a linear subspace shifted by a support vector.